by the Contractor. Construction practices shall include but are not limited to the following:

502.21 Prime Coat and Tack Coat – Applications of prime coat and tack coat are required and shall be applied unless otherwise directed by the Ministry Representative.

502.21.01 Surface and Weather Conditions – The liquid asphalt for prime coat or tack coat shall be applied when surface and weather conditions are favourable. The application of prime coat and tack coat shall meet manufacturer's requirements.

502.21.02 Spray Temperature – The liquid asphalt shall be sprayed within the temperature range specified by the supplier.

502.21.03 Prepared Granular Bases and Old Pavements – All prepared granular bases, old pavements, and portions of all other infrastructure that will be in contact with the new pavement paved shall be prime coated or tack coated at specified rates.

502.21.04 Application – The spray bars shall produce double coverage at one pass, with uniform spray and even pressure with application rates controlled to within ±25% of the specified application rate.

Spray bar nozzles shall be of the same type and size, set to produce uniformly fan-shaped sprays without atomization.

502.21.05 Spraying Faults – Any spraying faults shall be corrected by the Contractor.

502.21.06 Excess Liquid Asphalt – Excess liquid asphalt remaining unabsorbed shall be blinded with sand or fine aggregate at no expense to the Ministry.

502.21.07 Surface Condition – The surface to be tack coated shall be cleaned of dirt or other foreign material.

502.21.08 Application – Tack coat shall not be applied on sections of roadway longer than will be covered by one day of asphalt plant production.

502.21.09 Traffic – Traffic shall not be permitted on tack coat until it has cured and on prime coat until the primer has been absorbed into the granular surfacing.

Where it is not possible to keep the traffic off the treated surface, the surface shall be blinded with sand or fine aggregate, at no expense to the Ministry, prior to allowing traffic on surface.

502.21.10 Curing – Prime coats shall be allowed to cure for a minimum of 24 hours or to the satisfaction of the Ministry Representative prior to the placing of asphalt mix and tack coats shall be allowed to cure to a state that minimizes tracking prior to the placing of asphalt mix.

502.21.11 Maintenance – The Contractor shall, at the Contractor's expense, maintain the prime coat and/or tack coat.

502.22 Adjacent Mats, Joints, Edges, and Let-downs

502.22.01 Longitudinal Joints – Longitudinal joints in the Top Lift of asphalt pavement will only be permitted where lane dividing lines are to be painted. Longitudinal joints in a Lift shall be offset within 150 to 300 mm from joints in the underlying Lift. Joints shall be pinched as soon as practicable, using best compaction practices.

502.22.02 Longitudinal Edge – Any longitudinal edge that has been damaged by traffic or equipment shall be trimmed to provide a vertical abutting face. Vertical surfaces of roadway appurtenances shall be tack coated to the top of the new pavement only.

502.22.03 Contact Edge – The contact edge of abutting cooled asphalt pavement shall be thoroughly painted with a uniform coat of emulsified asphalt.

502.22.04 Paving Adjoining Mats – When paving an adjoining parallel mat there shall be at overlap onto the previously paved surface of 25 ±15 mm. This overlap shall be properly constructed to form a homogeneous bond between the two mats.

502.22.05 Disposal of Raking Material – Any raked material from the joint shall not be placed on the new mat or placed in or in front of the paver. The Contractor shall dispose of the material in a manner acceptable to the environmental agencies and the Ministry Representative.

502.22.06 Discontinued Paving – When paving is discontinued in any lane on a traffic bearing roadway, the asphalt pavement shall be tapered down at a minimum slope of 25H:1V. When paving resumes this letdown shall be cut back so as to form a vertical face that matches the required depth being paved. Material removed from the taper shall be disposed of in a manner acceptable to the environmental agencies and the Ministry Representative.

502.23 Stockpiling Aggregates

502.23.01 Stockpile Construction – Stockpiles shall be constructed as specified in SS 202.21.

502.23.02 Stockpiles for Different Materials – Stockpiles of different types of material shall be located and constructed in such a manner as to prevent intermingling of the types and to prevent segregation.

502.23.03 Stockpile Requirements Prior to Mix Production – Before plant mixing commences, the Contractor shall have in stockpile a minimum of 20,000 tonnes of asphalt mix aggregate or 50% of the total quantity of asphalt aggregate required for the mix, whichever quantity is greater. These quantities shall be maintained throughout the crushing period. These requirements may be waived by the Ministry Representative in circumstances where such stockpiles cannot be accommodated, such as where materials are being barged in or are being hauled to the plant from a remote site.

502.24 Mix Production

502.24.01 Mixing Temperature – The temperature of the asphalt mix measured at the plant discharge chute shall be maintained at ±15°C of the Design Mixing Temperature designated in the accepted Mix Design, with adjustments within that range made at the Contractor's discretion. Where the Contractor plans to adjust the actual mix temperature to 10°C or more above the Design Mixing Temperature, the Contractor shall notify the Ministry Representative prior to making the adjustment. To optimize mix properties during inclement weather or to address other specific circumstances, the Ministry Representative may agree, in advance, to a higher mixing temperature.

Mix produced at a temperature above the upper tolerance limit may be deemed Reject Mix by the Ministry Representative.

502.24.02 Residual Moisture Content – The residual moisture content of the asphalt mix prior to compaction shall not exceed 1% by mass. Should foaming or bubbling persist, even below the allowed percentage of moisture, the Contractor shall modify operations accordingly.

502.25 Placing the Asphalt Mix

502.25.01 Surface Condition – Asphalt mix shall only be placed on clean dry surfaces free from all foreign materials, and when weather and conditions are suitable. Normally, asphalt mix is only placed when the ambient air temperature is 5°C and rising and for Top Lift when the average surface temperature is also above 5°C.

502.25.02 Asphalt Levelling Course Requirements – Asphalt Levelling Course shall be placed in one or more Lifts, with a maximum thickness of 75 mm per Lift. The amount of Levelling Course placed shall not exceed the quantity listed in Schedule 7 unless otherwise authorized in writing in advance by the Ministry Representative.

All asphalt mix shall be paver laid, unless otherwise authorized by the Ministry Representative.

Any mix laid prior to the establishment of the Job Mix Formula (per SS 502.08) shall normally be placed in a location where performance of the mix is of lower concern, typically as level course, on a low volume side road, and/or in Bottom Lift. The location proposed by the Contractor shall be subject to the authorization of the Ministry Representative.

502.25.03 Lay-Down Operation – Asphalt mix should be delivered to the paver at a constant rate sufficient to allow continuous placement.

502.25.04 (not used)

502.25.05 Persons in the Vicinity of the Paver – When in the vicinity of an operating paver and in front of the screed, workers and inspectors shall stand in full view of the paver operator, clear of side arms, wings, and screed, and never directly in front of the machine hopper.

502.25.06 Vibrating Tampers and Hand Rollers – Along curbs, manholes, headers and similar structures which are not accessible to rollers, thorough compaction shall be obtained, by use of vibrating tampers and hand rollers or other suitable methods.

502.25.07 Traffic – Traffic will not be permitted on the finished pavement until it has cooled to such a temperature as to ensure that no deformation or flushing of the surface will occur.

502.25.08 Release Agents – Hydrocarbon-based release agents are not permitted. Other release agents may be used in accordance with their manufacturer's recommendations. Application rates of any release agent (including water) shall be limited to the minimum necessary, without any excess, to achieve the desired result.

502.25.09 Ponding – Areas of ponding will be noted as an obvious defect and remedial works will be required.

502.26 Shoulder Build-up – Shoulder build-up is typically required where an existing roadway is being overlaid with wider shoulders. It involves construction of base course aggregate (adjacent to a Lift of pavement) which will be overlaid by one or more Lifts of asphalt. Shoulder build-up shall be constructed:

(a) In accordance with the SS 202.26;

(b) To the lines and grades indicated in the Contract, generally to a finished compacted level that matches level and grade of the adjacent asphalt level; and

(c) With cross-fall that matches that of the finished asphalt surface above the shoulder build-up area.

502.27 Shouldering

502.27.01 Shouldering Aggregate – Shouldering aggregate shall be conform to SS 502.06.04.

502.27.02 Lines and Grades – The finished surface of the granular shoulder shall be true to grade and cross section, and be free from ruts, segregation or other irregularities.

502.27.03 Aggregate Segregation – All granular shoulder materials shall be handled in such a manner that segregation does not occur.

502.27.04 Moisture Adjustment – If necessary for compacting, the moisture content of the shoulder aggregate shall be adjusted through either drying or applying water. Addition of water is incidental.

502.27.05 Compaction – Shoulders shall be thoroughly compacted to a state that will not rut more than 5 mm under a pick-up truck steering axle wheel load.

Where the base under the shoulder has been constructed as part of the Work, the degree of compaction of the granular shoulder measured in accordance with ASTM D6938 shall be greater than 98% of the Standard Proctor maximum dry density determined in accordance with ASTM D698.

502.27.06 Adjacent Pavement Damage – If the adjacent asphalt pavement is damaged during the shouldering operation, the damage shall be repaired to the satisfaction of the Ministry Representative, at no expense to the Ministry.

502.27.07 Interim Shouldering – For safety reasons, when a roadway with gravel shoulders is used by the travelling public, an intermediate layer of shouldering aggregate shall be placed following final rolling of levelling course or multiple Lifts of asphalt pavement thicker than 50 mm, in advance of the main shouldering operation.

Should the Work extend past the Completion Date and be carried over to the next paving season suitable interim drainage control measures shall be put in place, all at no cost to the Ministry.

PAYMENT

502.30 General – Payment at the Unit Prices for the supply of primer/tack coat materials, the application of the spray primer/tack coat, the supply of paving aggregates into stockpile, the construction of asphalt pavement including the supply of asphalt cement, and for shouldering shall be full compensation for completing the supply and installation of asphalt pavement and shouldering on prepared surfaces in accordance with the contract requirements. Applicable payment adjustments (Additions or subtractions as applicable) shall be applied in accordance with the Payment Adjustment section of this Section.

Compensation for Quality Control shall be at the price bid for Quality Control or, in the absence of such a bid Item in Schedule 7 – Approximate Quantities and Unit Prices, incidental to the Work.

502.31 Acceptance at Adjusted Payment – Acceptance of any Lot at adjusted payment will occur if it complies with the requirements of the QC Plan and the sections on Equipment and Plant; and Construction of this Specification, contains no obvious defects as per SS 502.32, and if:

(a) Test results for EPS acceptance parameters (density, gradation, asphalt content, smoothness, segregation, and application rate) are such that the Lot meets the requirements for acceptance at a reduced payment;

(b) Lot is acceptable in respect of all other requirements; and

(c) Contractor has not notified the Ministry Representative in writing that it will exercise its option to either repair or remove and replace the work, at its own cost, with work meeting the requirements for acceptance at full or increased payment.

502.32 Rejection for Workmanship Defects – Work may be rejected if it does not comply with the requirements of the QC Plan and the Construction section of this Section.

Additionally, the finished surface of any Asphalt Pavement Lift shall have a uniform texture and be free of visible signs of poor workmanship. Any obvious defects as determined by the Ministry Representative such as, but not limited to the following, may be cause for automatic rejection of asphalt pavement regardless of the values of any other acceptance parameter:

(a) Individual bumps and dips that exceed 12 mm over 3 m;

(b) Areas of excess or insufficient asphalt;

(c) Improper matching of longitudinal and transverse joints;

(d) Roller marks or roller pick-ups;

(e) Excess tracking of prime or tack coat;

(f) Areas contaminated by fuel oil or other deleterious materials;

(g) Tire marks; or

(h) Cracking or tearing.

When asphalt pavement is rejected because of obvious defects, the minimum area of rejection will be the actual length of the defect for the full width of the driving lane in which the defect exists.

Rejected work shall be promptly repaired, remedied, overlaid, or removed and replaced in a manner acceptable to the Ministry Representative. The Contractor shall be responsible for all costs including materials.

No payment will be made for work in any Lot or Sub-Lot, which has been rejected, until the defects have been remedied.

502.33 Partial Payment for Rejected Work – In the Ministry Representative's sole discretion and without setting precedence, where any Work is reject but the Ministry Representative determines that it may be left in place, the Ministry Representative may authorize partial payment to the Contractor as full compensation for any residual value the Work may have. Notwithstanding the foregoing, the Ministry is under no obligation to make any payment for such Work.

502.34 Primer and Tack Coat

502.34.01 Supply of Primer and Tack Coat – Payment to Supply Primer and Tack Coat shall be at the Unit Price per litre for the quantity of material actually sprayed onto the roadway measured prior to any additional diluting by the Contractor where applicable. The measured quantity shall not be greater than the total of the bills of lading.

Such payment shall be full compensation for supplying, handling, storing, sampling and testing of the material and all other related work.

502.34.02 Application of Primer and Tack Coat – Payment for the application of Primer and Tack Coat will be at the Unit Price per litre of primer actually sprayed onto the roadway. In the case of an emulsion, the payment quantity shall include any water added by the Contractor with the authorization of the Ministry Representative.

Such payment shall be full compensation for diluting the material when required and spraying it on the surfaces to be paved, and for all incidental costs arising from priming and tack coating.

502.35 Paving Aggregate into Stockpile – Payment for Paving Aggregate into Stockpile shall be at the Unit Price per tonne for the quantity of paving aggregate placed into stockpile in accordance with the following.

As the aggregate is crushed into stockpile, progress payments will be made against the bid item, up to the quantity shown in the "Approximate Quantity" column of the Schedule 7 - Schedule of Approximate Quantities and Unit Prices.

The final payment quantity for paving aggregate quantity will be equal to the corresponding asphalt mix quantity actually laid, without any adjustment for AC or moisture content. As a result, the final quantities and payments may vary from those used in prior progress payments; down if the mix quantity underruns and up if it overruns.

Such payment shall be full compensation for all work including but not limited to the production, supply and stockpiling of all paving aggregates.

502.36 Asphalt Pavement – Payment for Asphalt Pavement constructed in place shall be at the Unit Price per tonne for the quantity of mix placed in accordance with the Contract requirements.

Subject to the exception noted below, only acceptable asphalt pavement will be included in the payment quantity.

Where overlays are used as a corrective measure, in accordance with SS 502.58:

- The overlay quantity will not be included in the payment quantity, but
- The quantity of asphalt pavement covered by the overlay will be included in the payment quantity, whether or not it was acceptable.

Such payment shall be compensation in full for all work including but not limited to: loading the aggregate into the feeders; drying the aggregate; supplying, metering, and adding the asphalt cement; mixing, loading, weighing, hauling, dumping, spreading, compacting and finishing the asphalt pavement.

502.37 Asphalt Mix Antistrip Additives

502.37.01 Where Specified in the Special Provisions – Where the Special Provisions direct the addition of an antistrip additive, all costs associated with supplying, storing, and blending the additive into the asphalt mixes shall be incidental to the prices bid for the varying classes of asphalt mix.

502.37.02 Where Not Specified in the Special Provisions – Where the Special Provisions do not direct the addition of an antistrip additive but the need was identified in accordance with SS 502.01.01, compensation for supplying, storing and blending the additive into the asphalt mixes shall be made at a negotiated price or on a Force Account Basis and any authorized payments will be made from the Provisional Sum for Site Modifications.

502.38 Asphalt Cement "Bump" – The Contractor will be compensated for additional oil ordered by the Ministry Representative in accordance with SS 502.08.09 at the AC cost FOB the asphalt plant, demonstrated by the AC supplier's invoice, with mark-up on a Force Account Basis for the material only.

Payment will be calculated in accordance with Equation 502-1, on a Lot-by-Lot basis based on the difference in asphalt cement content determined from Ministry Quality Assurance (QA) samples and that in the Contractor's production Job Mix Formula, to a maximum of the bump and a minimum of the Contractor's proposed Job Mix Formula AC content. No compensation will be made for additional AC in any rejected mix.

Equation 502-1: Payment for Asphalt "Bump"

$$\$_{\text{Lot n}} = 110\% \times \text{Cost}_{\text{AC}} \times t_{\text{Lot n}} \times \left| \left(\frac{AC_{Burn}}{100 + AC_{Burn}} \right) - \left(\frac{AC_{JMF}}{100 + AC_{JMF}} \right) \right|$$

Where:

$\$_{Lot\ n}$	*= Compensation due for additional AC in Lot "n"*
$Cost_{AC}$	*= Supplier's invoiced cost for AC, dollars per tonne*
$t_{Lot\ n}$	*= Tonnes of asphalt mix laid and accepted in Lot "n", including AC*
AC_{Burn}	*= Asphalt cement content (% dry mix) of Lot determined from Ministry QA samples, to a maximum of the bumped AC content and a minimum of AC_{JMF}*
AC_{JMF}	*= Asphalt cement content (% dry mix) of Contractor's production Job Mix Formula before the bump*
Note:	*110% includes the percentage markup for materials paid on Force Account*

502.39 Payment for Rejected Work Made Acceptable – When defects have been remedied in Lots or Sub-Lots which had been rejected, payment for the original quantity of material in those Lots or Sub-Lots will be made subject to payment adjustments and penalty assessments and subject to SS 502.58.

No payment adjustment will be made for any material used to replace, repair or overlay rejected work and all corrective work shall be performed entirely at the Contractor's expense.

502.40 Shouldering – Payment for Shouldering will be at the Unit Price bid per tonne or cubic metre (to the neat lines), whichever is specified in the Schedule 7 - Schedule of Approximate Quantities and Unit Prices. The price bid shall be full compensation for supplying, hauling, placing, moisture adjustment, and compaction of the aggregate to the required grades and crossfall.

502.41 Surplus Aggregate

502.41.01 Surplus Aggregate in Private Pits – At the discretion of the Ministry Representative, the Ministry may or may not purchase surplus aggregate in stockpile in a private pit. Generally, the Ministry will provide payment for the processing costs of surplus aggregate in stockpile in Ministry Pits only, and only to a limited quantity. However, should the Contractor produce surplus aggregate in a private pit, and the Ministry intends to purchase these surplus aggregates, the Contractor shall be required to provide a written agreement with the owner of the property. This document shall indicate that the Ministry will have free access to and use of the surplus aggregate in stockpile for a period of 12 months after the completion of the contract work. If the Contractor undertakes private work from within the private pit, measurements for surplus aggregate in stockpile will not be taken until the completion of the private work, ensuring that the Ministry does not pay for aggregate used on private works. All surplus aggregate shall be properly stockpiled.

502.41.02 Ministry Purchase of Surplus Aggregate – Should the Ministry proceed with the purchase of surplus aggregate, upon completion of the contract, the Ministry will purchase surplus paving mix aggregate as indicated herein.

The provisions below apply only to AMA, including AMA used as shouldering aggregate. All other surplus aggregates, including grading aggregates used as shouldering surfacing, base and subbase aggregates will be paid in accordance with SS 202.

If the quantity of Asphalt Pavement actually incorporated into the works is less than the estimated quantity, as stated in the Schedule 7 - Schedule of Approximate Quantities and Unit Prices, the Ministry will purchase aggregate up to 100% of the quantity required by the contract, to be paid under the appropriate Unit Price Item for Aggregate in Stockpile. The surplus aggregate must, when singly or combined, meet the gradation requirements set out in the Job Mix Formula.

The portion of material remaining in a stockpile eligible for compensation as surplus aggregate shall be the lesser of:

(a) The quantity of mix as stated in the Schedule 7 - Schedule of Approximate Quantities and Unit Prices minus the quantity of aggregate actually incorporated in the works, and

(b) The volume of the remaining stockpile as determined under SS 502.41.03.

No payment shall be made for any surplus shoulder and paving aggregate remaining at contract completion in excess of the contract requirement. Material remaining in a Ministry pit is the property of the Ministry.

502.41.03 Measurement of Surplus Aggregate – The Ministry will determine the volume of aggregate by surveying the stockpile(s) using string-line techniques, and determine volume using prismoidal volumes between surfaces. Alternatively, the Ministry Representative may elect to survey using cross-section techniques and/or determine volumes using end-area volumes.

The base of the pile will be determined from a pre-stockpile survey or, where such a survey is not available, from a surface determined by the Ministry Representative as being a reasonable interpolation from the intersection of the pile sides with the adjacent ground level. Where there are two or more stockpiles of aggregate meeting the same gradation classification, the compensation will be based on the cumulative quantity of those stockpiles.

Where necessary, stockpile volume shall be converted to mass using the conversion factor of 2.0 tonnes per cubic metre.

PAYMENT ADJUSTMENTS

502.50 Payment Adjustment – Payment adjustments resulting from the application of this Section will be effected on each progress payment as follows.

Notwithstanding any other provision of the Contract, including full or partial waivers of payments adjustments for any or all of the following five rated characteristics:

- density;
- AC content;
- gradation;
- application rate; and
- segregation

the rejection limits for each of those characteristics will apply to the Work.

For smoothness, the payment adjustments and rejection limits shall apply as specified in SS 502.57 Smoothness.

502.50.01 Density, Asphalt Content, and Gradation – For each Lot paid for by the tonne (t), the applicable payment adjustment derived from Table 502-I or Table 502-J (Density), Table 502-K (Asphalt Content), and Table 502-L (Gradation), in dollars per tonne, will be

expressed as positive in the case of increases and negative in the case of decreases.

The algebraic sum of these unit adjustments will then be applied to the payment quantity for the Lot. The resulting amount, in dollars, will be the net payment adjustment, positive or negative, for that Lot.

502.50.02 Application Rate – Payment Adjustments from Table 502-M (Application Rate) shall be computed for the Lot.

The algebraic sum of the net payment adjustments for all such Lots for which payment is authorized on the current progress payment, computed in dollars, shall be the total payment adjustment for density, asphalt content, aggregate gradation, and material application rate for the current progress payment.

502.50.03 Segregation and Smoothness – For each Lot with payment adjustments based on kilometre (km), the applicable unit adjustment derived from Table 502-O (Segregation) and Table 502-P (Smoothness), in dollars per Lot ($/lane km), will be expressed as positive in the case of increases and negative in the case of decreases, and will be the payment adjustment, positive or negative, for that Lot.

The algebraic sum of the payment adjustments for all such Lots for which payment is authorized on the current progress payment, computed in dollars, shall be the total payment adjustment for segregation and smoothness for the current progress payment.

502.50.04 Total Payment Adjustment – The algebraic sum of the total payment adjustments for density, asphalt content, aggregate gradation, material application rate, segregation and smoothness, derived in accordance with SS 502.50.01 through SS 502.50.03 above, shall be the total payment adjustment, positive or negative, in dollars, for all attributes for the current progress payment.

The total payment adjustment will be made by a single entry in computing the current progress payment.

502.50.05 Progress Payments – The process set out in SS 502.50.01 through SS 502.50.04 above will be used in computing each progress payment to which it is applicable.

502.51 Initial 1,000 tonnes of Mix – For the first 1,000 tonnes of asphalt mix produced under a Contract, the following provisions take precedence over all other payment and payment adjustment provisions of SS 502.52 through SS 502.57 inclusive, but do not take precedence over the rejection criteria.

502.51.01 Payment Adjustments – Unless requested otherwise by the Contractor in writing in advance of mix production, the first 1,000 tonnes of asphalt mix production and placement shall not be subject to the bonus/penalty payment adjustments for AC content, density and gradation. Payment adjustments will apply to smoothness, segregation and application rate if the mix is applied in a Top Lift location.

502.51.02 Acceptance Limits – Contrary to any other provision of this SS 502, any mix produced during the initial 1,000 tonnes will only be considered acceptable if:

(a) Asphalt Mix Design has been accepted by the Ministry Representative in accordance with SS 502.08.05;

(b) Aggregate gradation per SS 502.08.02 is within the gradation limits specified in Table 502-D or the banana formed by applying the Table 502-L, column 3 variation limits to the JMF gradation;

(c) All other properties fall inside the allowable limits specified, in this SS 502 and the Special Provisions, for the mix, including

(i) For Marshall mixes, Table 502-E; and

(ii) For Superpave mixes, Table 505–F.

(d) AC content of the sample is within ±0.5% of the design value for top lift and is within ±0.55% of the design value for the bottom lift.

All values are measured against the Asphalt Mix Design value at the time the sampled mix was produced.

Any mix with any characteristic outside the above limits is Reject Mix.

Additionally, rejection limits for smoothness, segregation, density and application rate shall apply in accordance with this SS 502 and the Special Provisions.

502.52 Density

502.52.01 Lot – A Lot for density shall be one day's scheduled production of at least 7 hours plant production where no changes have occurred to criteria such as but not limited to:

(a) Accepted Job Mix Formula;

(b) The specific Lift that is being placed; and

(c) The required material application rate.

A change in any above criteria may require a new Lot designation.

Where one day's production is less than 7 hours, the material will be added to the next Lot that has the same criteria, as described above, except that if a test indicates that this production is subject to a payment adjustment or to rejection, or if no further material will be produced with the same criteria, this production will be designated as a separate Lot.

A Lot shall be no more than two days total production even if the above criteria have not changed or been met.

502.52.02 Sub-Lot – For density, a Lot shall be divided into three equal Sub-Lots, defined by lineal metres of production

502.52.03 Coring – The Contractor shall be responsible for providing all core samples for quality assurance and payment adjustment purposes. The randomly selected locations for the cores shall be supplied by the Ministry Representative to the Contractor. The Contractor shall provide cores sized in accordance with Table 502-H. Coring shall occur in the presence of a Ministry representative, within two days after the Lot has been completed and prior to application of any subsequent lift. Unless otherwise directed by the Ministry Representative, the Contractor shall prepare the cores prior to the submission by removing all material not representative of the pavement Lift to be tested.

The Contractor shall immediately provide all cores taken (including those damaged during extraction) to the Ministry Representative or delegate thereof.

The Contractor shall fill all core holes before the roadway is re-opened to traffic. Core holes shall be filled by the following method:

(a) Empty the hole of water and loose material.

(b) Remove any excess moisture by wiping the inside with a dry towel.

(c) Tack coat the inside surfaces and the outside perimeter with an emulsified asphalt.

(d) Place asphalt mix in loosely, so that the compacted Lifts do not exceed 75 mm.

(e) With a minimum of 20 blows per Lift, compact the loose material using a minimum 2 kg sledgehammer and tamper.

(f) For additional Lifts repeat Steps (a) to (e).

(g) The final Lift shall be a minimum thickness of 25 mm and finished to a level higher but not exceeding 6 mm, than the elevation of the surrounding pavement.

Table 502-H: Test Sample Size and Source Locations

Test	Sample Source
Density	150 mm dia. road cores for Superpave, Coarse Mix, and any other product with maximum aggregate size greater than 25 mm. 100 mm dia. road cores for all other mixes
AC Content	Loose Mix Samples
Smoothness	Centre of the Lane
Aggregate Gradation	Loose Mix Samples
Segregation	Roadway Pavement

The Contractor may use an alternative method if acceptable to the Ministry Representative.

All costs associated with obtaining the cores, including the filling and compaction of the core holes are considered incidental to the Contract and are the responsibility of the Contractor.

502.52.04 Percent Density – The average in-place density will be determined from core samples of the completed Lift of pavement.

One random core sample will be obtained from each Sub-Lot and the test results for the three Sub-Lots will be averaged to determine the percent density for the Lot.

(a) Marshall Mixes – For the determination of Marshall Briquette Density, the Contractor shall obtain the samples as outlined in APPENDIX 502-B for each Sub-Lot. All costs associated with sampling, shall be the Contractor's responsibility.

From each Sub-Lot sample, the Contractor will form three briquettes as per the procedure identified in Table 502-A and tested, and the ASTM D2726 test results will be averaged to obtain the test value for that sample.

The Marshall Briquette Density for a Lot is the average of the test results from the three Sub-Lots. The Marshall Briquette Density value for the Lot shall be provided to the Ministry Representative prior to the provision of the coring locations to the Contractor.

Equation 502-2: Marshall Percent Density

$$\text{Marshall Percent Density} = \left(\frac{\text{In-place Density of Sample}}{\text{Marshall Briquette Density}} \right) \times 100$$

At the discretion of the Ministry Representative, the Ministry may at any time, perform and use the Ministry results in accordance with procedures as outlined above, to obtain the value for Marshall Briquette Density which will replace the values achieved under Quality Control Testing.

Should the initial test results in any Sub-Lot be lower than 97.0%, one additional core will be taken and tested, and that result shall be averaged with the initial result to determine the new percent density for the Sub-Lot.

In addition, the Contractor shall determine the individual Maximum Theoretical Densities (MTD) per ASTM D2041 for each of the Sub-Lot samples and average the results of them to obtain the Lot MTD. The Contractor shall report the Lot MTD to the Ministry Representative prior to the provision of the coring locations to the Contractor.

(b) Superpave Mixes – The Contractor shall determine the individual Maximum Theoretical Densities (MTD) per ASTM D2041 for each of the Sub-Lot samples and

average the results of them to obtain the Lot MTD. The Contractor shall report the Lot MTD to the Ministry Representative prior to the provision of the coring locations to the Contractor.

At the discretion of the Ministry Representative, the Ministry may at any time use the same procedures to obtain the values for Maximum Theoretical Density which will replace the values determined by the Contractor.

The average in-place density will be determined from core samples of the completed Lift of pavement. Cores shall be sized in accordance with Table 502-H.

Equation 502-3: Superpave Percent Density

$$\text{Superpave Percent Density} = \left(\frac{\text{In-place Density of Sample}}{\text{Lot MTD (ASTM D 2041)}} \right) \times 100$$

The Maximum Theoretical Density for a Lot is the value calculated, defined as the average of the test results, from the three Sub-Lots. The Maximum Theoretical Density value for the Lot shall be provided to the Ministry Representative prior to the provision of the coring locations to the Contractor.

Should the initial test results in any Sub-Lot be lower than 92.0%, one additional core will be taken and tested, and that result shall be averaged with the initial result to determine the new percent density for the Sub-Lot.

502.52.05 Payment Adjustments – The payment adjustment for percent density will be the amount shown in Table 502-I or Table 502-J, as applicable, for the Sample Mean of the test results for the Lot.

502.52.06 Rejection Limit – The rejection limit for percent density is the limiting value of the Sample Mean as shown in Table 502-I or Table 502-J, as applicable. If any Sub-Lot has a value within the reject zone, the Sub-Lot is rejected and not paid for.

Acceptance or rejection will be determined for each Sub-Lot. One opportunity will be provided to the Contractor, using non-destructive test methods, to isolate the area of low density and conduct additional compaction providing no damage is incurred to the new pavement. The Contractor shall mark out the extents of the area identified using chalk or other temporary marking acceptable to the Ministry Representative. One additional core will be taken randomly in the area of low density and will replace the core with previous low density and a new value will be calculated. The Ministry Representative shall be advised as to the scheduling of the re-compaction effort prior to the work taking place.

Where any Marshall Mix core density is less than 96% (91% for Superpave mixes), the Contractor shall either overlay or remove and replace the previously placed area of reject mix. If the test result for density of a Sub-Lot is outside the acceptance limits, the Sub-Lot is rejected automatically regardless of the values of other acceptance parameters. To minimize the cost of rejection to the Contractor, the Contractor will be provided the opportunity to isolate the area of low density within the Sub-Lot. The limits of the low-density area must be verified and accepted by the Ministry Representative before remedial work proceeds.

502.52.07 Payment Adjustment for Density Rejected Work Made Acceptable – The payment adjustment for density will be based on testing of the replacement or overlay material where applicable. Where replacement or overlay material does not cover the entire Lot or Sub-Lot, prior tests of the uncovered area will be averaged with new tests on the corrective work

Table 502-I: Payment Adjustments for Density – Marshall

Marshall % Density Lot Average	Payment Adjustment ($ per tonne)
≥ 98.5	+$2.00
≥ 98.0 to < 98.5	+$1.50
≥ 97.5 to <98.0	+$1.00
≥ 97.0 to <97.5	+$0.50
≥ 96.5 to <97.0	-$1.00
≥ 96.0 to <96.5	-$2.00
<96.0	**REJECT**

Table 502-J: Payment Adjustments for Density - Superpave

Superpave % Density Lot Average	Payment Adjustment ($ per tonne)
≥ 95.0	+$2.00
≥ 94.0 to <95.0	+$1.50
≥ 93.0 to <94.0	+$1.00
≥ 92.0 to <93.0	+$0.50
≥ 91.6 to <92.0	-$1.00
≥ 91.0 to <91.6	-$2.00
<91.0	**REJECT**

502.53 Asphalt Content

502.53.01 Lot – A Lot for AC content shall be one day's scheduled production of at least 7 hours plant production where no changes have occurred to criteria such as but not limited to:

(a) Accepted Job Mix Formula;

(b) The specific Lift that is being placed; and

(c) The required material application rate.

A change in any above criteria may require a new Lot designation.

Where one day's production is less than 7 hours the material will be added to the next Lot that has the same criteria, as described above, except that if a test indicates that this production is subject to a payment adjustment or to rejection, or if no further material will be produced with the same criteria, this production will be designated as a separate Lot.

A Lot shall be no more than two days total production even if the above criteria has not changed or been met.

502.53.02 Sub-Lot – For AC content, a Lot shall be divided into three equal Sub-Lots, defined by lineal metres of production.

502.53.03 Determination of Asphalt Content – The asphalt content of the asphalt mix will be determined from loose mix samples obtained from each Sub-Lot (3 per Lot) and tested in accordance with the ASTM D6307 test procedures listed in Table 502-A.

One random sample will be obtained from each Sub-Lot as per APPENDIX 502-B. The asphalt content of the asphalt mix will be determined using the average of the results obtained from all of the Sub-Lot samples. The actual Asphalt Content of the Lot will be compared to the Job Mix Formula Asphalt Content and the deviation will be used for payment adjustment purposes.

The ignition oven correlation factor will be applied to the AC Content measured by the ignition oven, and the corrected AC Content used to determine acceptability of the mix and any payment adjustments.

502.53.04 Payment Adjustments – The payment adjustment for asphalt content will be the amount shown in Table 505–F, where deviation in AC Content is determined by the percent difference, based on kg of AC per 100 kg of dry aggregate, between the JMF AC content and the actual AC Content.

However, if a Lot contains any Sub-Lot that falls within the reject zone of Table 502-K:

(a) no bonus payment will be made for the Lot; and

(b) any penalty for the Lot will be based on the average AC content of all Sub-Lots, including those within the reject zone.

502.53.05 Rejection Limit – Rejection limits for asphalt content are the limiting values of the Sample Mean as shown in Table 502-K beyond which the Lot is rejected and not paid for.

If the asphalt content of a Lot is within the reject zone, the Lot is rejected automatically regardless of the values of other acceptance parameters.

For Top Lift deviations of more than 0.50%, the Contractor shall either overlay or remove and replace the Lot.

For lower Lift deviations of more than 0.55%, no payment will be made, and the Ministry Representative will determine whether removal and replacement is necessary.

Table 502-K: Payment Adjustments for Deviation of Asphalt Content (AC)

Differences of Actual AC Content from Designed AC Content Specified in JMF (AC in %)		
Deviation from Asphalt Mix Design JMF	**Payment Adjustment $ per tonne**	
	Top Lift	**Lower Lifts**
-0.56 or less	**REJECT**	**REJECT**
-0.55 to -0.51	**REJECT**	-9.00
-0.50 to -0.46	-8.00	-8.00
-0.45 to -0.41	-7.00	-7.00
-0.40 to -0.36	-5.00	-5.00
-0.35 to -0.26	-3.00	-3.00
-0.25 to -0.16	-1.00	-1.00
-0.15 to -0.01	0.00	0.00
0.00 to +0.15	+2.00	+2.00
+0.16 to +0.30	+1.50	+1.50
+0.31 to +0.35	0.00	0.00
+0.36 to +0.40	-2.00	-2.00
+0.41 to +0.45	-3.50	-3.50
+0.46 to +0.50	-5.00	-5.00
+0.51 to +0.55	**REJECT**	-6.50
+0.56 or greater	**REJECT**	**REJECT**

502.53.06 Payment Adjustment for Asphalt Cement Content Rejected Work Made Acceptable – The payment adjustment for Asphalt Content will be based on testing of the replacement or overlay material where applicable. Where replacement or overlay material does not cover the entire Lot or Sub-Lot, prior tests of the uncovered area will be averaged with new tests on the corrective work.

502.54 Aggregate Gradation

502.54.01 Lot – A Lot for aggregate gradation shall be one day's scheduled production of at least 7 hours plant production where no changes have occurred to the criteria such as but not limited to:

(a) Accepted Job Mix Formula;

(b) The specific Lift that is being placed; and

(c) The required material application rate.

A change in any of the above criteria may require a new Lot designation.

Where one day's production is less than seven (7) hours the material will be added to the next Lot that has the same criteria, as described above, except that if a test indicates that this production is subject to a payment adjustment or to rejection, or if no further material will be produced with the same criteria, this production will be designated as a separate Lot.

A Lot shall be no more than two days total production even if the above criteria has not changed or been met.

502.54.02 Sub-Lot – For aggregate gradation, a Lot shall be divided into three equal Sub-Lots, defined by lineal metres of production.

502.54.03 Determination of Aggregate Gradation – Table 502-D, Aggregate Gradation Limits, specifies the aggregate gradation limits, which asphalt mix design, Job Mix Formulas, and field adjustments made in accordance with SS 502.08.10 must be based.

Table 502-L shows the limiting values for acceptance and for the determination of payment adjustments. These limits are expressed as departures, on each specified sieve size, from the percentage of material passing that sieve according to the Job Mix Formula. Aggregate gradation will be determined for each Lot from loose mix samples of the completed Lift of pavement. One random sample will be obtained from each Sub-Lot, and the aggregate gradation for the Lot will be determined by using the average of the 3 Sub-Lot tests to determine the Sample Mean.

Although the JMF design gradation must at all times be maintained within the limits specified in the Aggregate Table 502-D, the gradation of the mix may vary outside those limits, subject to SS 502.54.04.

502.54.04 Payment Adjustments – Where the Sample Mean for every specified sieve size falls within the limiting values prescribed in a particular column of Table 502-L the Lot will be accepted with a payment adjustment as indicated at the bottom of that column.

However, if a Lot contains any Sub-Lot that falls within the reject zone of Table 502-L:

(a) no bonus payment will be made for the Lot; and

(b) any penalty for the Lot will be based on the Sample Mean of all Sub-Lots, including those within the reject zone.

502.54.05 Rejection Limit – Where one or more values of the Sample Mean for the specified sieves falls outside the limiting value specified in Table 502-L, Column 3, the Lot is reject and the Contractor shall either overlay or, remove and replace the previously placed mix. The Lot will be assessed to determine whether it may remain in place, with payment made in accordance with SS 502.33.

Table 502-L: Payment Adjustments for Aggregated Gradation

Sieve Size mm	Divergence from JMF Grading Curve Percentage Passing by Mass (ASTM C117 and ASTM C136)		
	Column 1	Column 2	Column 3
12.5	±3.5	±4.5	±7.0
4.75	±3.0	±4.5	±6.0
0.600	±2.0	±3.5	±4.0
0.075	±0.75	±1.0	±1.5
Payment Adjustment $ per tonne	**+$0.75**	**$0.00**	**-$1.50**

502.55 Material Application Rate

502.55.01 Lot – A Lot for material application rate shall be one day's scheduled production of at least 7 hours plant production where no changes have occurred to the criteria such as but not limited to:

- The specific Lift that is being placed,
- The required material application rate.

A change in any above criteria may require a new Lot designation.

When one day's production is less than 7 hours, the material will be added to the next Lot that has the same criteria, as described above, except that if a test indicates that this production is subject to a payment adjustment or to rejection, or if no further material will be produced with the same criteria, this production will be designated as a separate Lot.

A Lot shall be no more than two days total production even if the above criteria has not changed or been met.

502.55.02 Sub-Lot – Sub-Lots are not used for material application rate.

502.55.03 Determination of Material Application Rate – Asphalt mix shall be applied to the roadway at the rate or rates specified in the Special Provisions, on the Drawings, at a corresponding mat thickness determined by dividing the design mix density by the specified application rate, or as otherwise directed by the Ministry Representative. Material Application Rates will be determined by the tonnage delivered to the paver as recorded by weigh tickets generated by automated scales, divided by the area covered by the Lot after allowance has been made for entrance letdowns and/or intersections. The Contractor shall provide the material application rate calculations to the Ministry Representative at the end of each shift.

502.55.04 Payment Adjustments – The payment adjustment for material application rate will be applied as

shown in Table 502-M, based on the actual material application rate, expressed as a percentage of the specified material application rate.

502.55.05 Rejection Limits – Where actual application rate for the Lot is within the reject zone as shown in Table 502-M, the Lot is reject.

A Lot rejected for application rate may be corrected by mill-and-fill or rejected with no remedial work required, subject to the authorization of the Ministry Representative.

Table 502-M: Payment Adjustments for Material Application Rate

Actual Application Rate (Percent of specified rate)	Payment Adjustment $ per tonne of material in the Lot (unless otherwise noted)	
	Bottom Lift or Single Lift	Top Lift of Multiple Lifts
≥ 110	-$7.00 for all material in the Lot up to 110% and no payment for product in excess of 110.0%	-$7.00 for all material in the Lot up to 106% and no payment for product in excess of 106.0%
≥ 106.0 to <110.0	-$5.00	
≥ 105.0 to <106.0		-$5.00
≥ 103.0 to <105.0	-$2.00	-$2.00
≥ 102.0 to <103.0	-$0.00	-$0.00
≥ 97.0 to <102.0	+$1.00	+$1.00
≥ 94.0 to <97.0	-$2.00	-$2.00
≥ 92.0 to <94.0	-$3.00	-$3.00
≥ 90.0 to <92.0	-$4.00	-$4.00
≥ 85.0 to <90.0	-$7.00	-$7.00
<85.0	**REJECT**	**REJECT**

502.56 Surface Segregation

502.56.01 Lot – A Lot shall be one kilometre length of Top Lift pavement for each driving lane.

502.56.02 Sub-Lot – Sub-Lots are not used for segregation.

502.56.03 Determination of Surface Segregation – The finished pavement shall be homogeneous, free from segregation and shall be uniform with respect to surface texture. A segregated area is defined as an area within the driving lanes of the pavement wherein the texture differs visually from the texture of the surrounding pavement.

Due to the nature of variation in asphalt mixes and their compactive characteristics, the definition of degrees of segregation will be established on a project-by-project basis. The Contractor and the Ministry Representative shall establish using photographs or other mutually agreed upon tools, the definition of slight, moderate and severe segregation. The Ministry Representative and the Contractor shall evaluate the first two Lots upon completion of the second Lot, after opening to traffic, to confirm the "agreed to" guidelines.

The Ministry Representative, with the Contractor's Representative, will observe the finished pavement to evaluate the existence, severity and extent of segregation and other surface defects. The evaluation will be completed prior to the issuance of the Completion Certificate.

The rating system for categorizing the severity of Top Lift surface segregation, and repair methodology is shown in Table 502-N.

502.56.04 Payment Adjustments – The payment adjustment for Top Lift segregation will be the applicable amount shown in Table 502-O.

Table 502-N: Segregation – Top Lift Only – Remediation Methodology

Segregation Severity	Visual Appearance	Repair Procedures
None	Uniform surface texture.	N/A
Slight	Matrix of asphalt binder, coarse and fine aggregate exists; visually increased presence of stone sizes.	Sand cement slurry.
Moderate	Significantly more stone than surrounding pavement; matrix of asphalt binder and coated sand particles is reduced.	Seal coat or sand cement slurry patch or neat hot mix patch or mill and fill patch.
Severe	Appears as an area of very stony mix - stone against stone - little or no matrix.	Remove and replace or overlay to limits authorized by Ministry Representative.

Table 502-O: Segregation – Top Lift Only – Payment Adjustments

Payment Adjustment	Number of Segregated Areas, by Category		
	Slight	Moderate	Severe
+$1,500 per lane km. *(Applies only if the criteria in all three adjacent columns are achieved)*	0 to 3	0	0
+$1000 per lane km. *(Applies only if the criteria in all three adjacent columns are achieved)*	4 to 6	0	0
No payment adjustment	7 to 9	0 to 1	0
-$500 per lane km. *(Applies if the criteria in any one or more of the adjacent columns occurs)*	10 to 12	2 to 4	0
-$1,000 per lane km. *(Applies if the criteria in any one or more of the adjacent columns occurs)*	13 to 16	5 or 6	0 to 2
-$3,000 per Lane km *(Applies if the criteria in any one or more of the adjacent columns occurs)*	17 or more	7 or more	3 or more

502.56.05 Repair – On Top Lift all segregation, including any areas outside the driving lanes assessed for the payment adjustment, shall be repaired according to Table 502-N.

All segregation patch repairs shall be completed to a rectangular shape.

Repair shall be to the neat lines and dimensions of the segregated area using sand cement slurry or other product acceptable to the Ministry Representative.

Acceptable sand cement slurry can be made as follows, with proportions varied as needed for workability:

- 25 litres of SS-1 (or equivalent) emulsion
- 4 – 5 kg (2 shovels) of ≤ 3 mm sand
- 2 – 3 kg (1 shovel) Type GU (general use) or Type GUL (General Use Limestone) Portland cement
- Additional water, if needed for workability.
- Or other products approved

After repairs, the Lot will be re-evaluated for acceptance, but not for segregations payment adjustments.

The Contractor shall be responsible for all costs associated with the repair of segregated areas.

502.57 Smoothness

502.57.01 Lot – A Lot for smoothness shall be one kilometre length of Top Lift pavement for each driving lane.

502.57.02 Sub-Lot – A Sub-Lot for smoothness shall be a 100 metre section of a Lot, or for the last Sub-Lot within a Lot, a section of up to 100 metres.

502.57.03 Traffic Control for Pavement Smoothness Testing – The Contractor shall provide traffic control for smoothness testing.

The Ministry Representative will provide sufficient notice to the Contractor regarding when the Ministry will perform smoothness testing. As directed by the Ministry Representative, the Contractor will be responsible to schedule and provide traffic control for the Ministry's smoothness testing, including a Shadow Vehicle and all signage necessary. The Contractor shall also be responsible for sweeping and any other preparation work required for smoothness testing. No extra payment will be made for sweeping or other preparation work; and the provision of traffic control for smoothness testing will also be considered incidental and no extra payment will be made.

The Contractor may also be required to provide an additional Shadow Vehicle or traffic control for smoothness testing as determined by the Ministry Representative. Payment for the provisions of an additional Shadow Vehicle or traffic control, if required, will be made under the Provisional Sum Item for Site Modifications (Schedule 7).

502.57.04 Determination of Pavement Smoothness – The finished pavement surface shall be tested by the Ministry using a Class I precision rolling profile measuring instrument, to determine the longitudinal profile and compute the International Roughness Index (IRI) in each driving lane. Profiles shall be measured and the IRI calculated in the centre of the lane for each Sub-Lot.

For any Sub-Lot between 50 m and 100 m in length, the IRI value shall be considered representative of a complete Sub-Lot. For any Sub-Lot less than 50 m in length, the IRI value will be combined with the proceeding Sub-Lot IRI value.

The profile shall be measured over the entire length of the pavement exclusive of structures and shoulder areas. Acceleration, deceleration and turning lanes are considered part of the driving lanes and shall be tested in accordance with this provision.

The following areas will be excluded from Smoothness EPS payment adjustments and reject limits:

- Areas with a speed limit less than 70 km;
- Curb and gutter;
- Ramps, acceleration or deceleration lanes less than 300 m in length;

- Freeway cloverleafs;
- Turn lanes and storage lanes; and
- Other areas as specified in the Contract.

The areas outlined above are not exempt from the requirements of SS 502.57.09 Smoothness Deficiencies.

For the measuring process, the Contractor shall provide the Ministry Representative a chalk guideline in the centre of the lane immediately prior to measurement.

502.57.05 Auxiliary Lanes – For smoothness testing, sections of the driving lanes that do not fall within the continuous through lanes, such as acceleration lanes, deceleration lanes and turning lanes, and lanes which are less than 1 km in length, shall be treated as follows. The ratio of the section length to the standard Lot length of 1 km shall be determined and the payment adjustment shall be pro-rated on this basis as in the following example:

For a segment 565 metres long, the payment adjustment factor is

= Length of segment / Length of standard Lot

= 565 m/1000 m

= 0.565

Hence the applicable payment adjustment is 0.565 times the payment adjustment for a 1 km Lot as determined from Table 502-P.

502.57.06 Sub-Lot and Lot Acceptance Limits – The reject limit and payment adjustments applicable to a road for smoothness are specified in Table 502-P. The default values in the "Standard" column shall apply except where the Special Provisions specify that "Alternate" values apply.

The International Roughness Index (IRI) value, calculated for each Sub-Lot, will be used to determine if the Sub-Lot or Lot will be accepted, and if so whether it will be subject to any payment adjustment.

The Lot IRI is the average of the individual IRI values for the Sub-Lots within the Lot.

A Sub-Lot is rejected if:

(a) It has an IRI within the applicable reject zone shown in Table 502-P, subject to SS 502.57.08;

(b) For Top Lift only, there are obvious defects per SS 502.32 or it has unrepaired smoothness deficiencies which require remediation in accordance with SS 502.57.09.

A Lot is rejected for Smoothness if any Sub-Lot is rejected.

502.57.07 Payment Adjustments – For Top Lift only, the payment adjustments per the applicable column of Table 502-P shall apply to each Lot.

Table 502-P: Payment Adjustments for Smoothness

Lot IRI (m/km)	Payment Adjustment Per Lot	
	Standard	**Alternate**
≤ 0.80	+$2,000	+$2,000
> 0.80 to ≤ 0.90	+$1,000	+$2,000
> 0.90 to ≤ 1.00	+$500	+$2,000
> 1.00 to ≤ 1.10	+$200	+$1,000
> 1.10 to ≤ 1.20	0	+$500
> 1.20 to ≤ 1.30	-$100	+$250
> 1.30 to ≤ 1.40	-$250	0
> 1.40 to ≤ 1.50	-$600	-$300
> 1.50 to ≤ 1.60	-$1,400	-$300
> 1.60 to ≤ 1.70	-$2,000	-$750
> 1.70 to ≤ 1.80	-$3,000	-$750
> 1.80 to ≤ 1.90	**REJECT**	-$1,500
> 1.90 to ≤ 2.00	**REJECT**	-$2,500
>2.00	**REJECT**	**REJECT**

502.57.08 Remedial Work – If the test results on a Sub-Lot of pavement indicate a payment reduction or rejection because of smoothness, the Contractor may propose remedial work to improve the smoothness. Such proposals are subject to the acceptance of the Ministry Representative, but such acceptance does not imply that the proposed remedy will be successful and does not reduce the Contractor's responsibility for meeting the acceptance requirements. Grinding may be acceptable, but an overlay may be required. Only one attempt may be made to improve smoothness, and this must be completed within fourteen (14) days from the time the Contractor receives notification from the Ministry Representative of the original smoothness test results for that Sub-Lot.

Following any attempt to improve the smoothness of a Sub-Lot or Sub-Lots, the Ministry Representative will retest the Sub-Lot or Sub-Lots, and the new results will replace the previous data for the purposes of determining acceptance and payment.

No payment will be made for any material, equipment or manpower used to improve, or attempt to improve, smoothness.

502.57.09 Smoothness Deficiencies – Smoothness deficiencies (bumps and dips) less than 12 mm over 3 m will not have remedial work required. Individual deficiencies between 8 mm and 12 mm over 3 m will result in a $200.00 penalty for each occurrence. Deficiencies exceeding 12 mm over 3 m will require remedial work.

502.58 Overlays as a Corrective Measure – If an overlay is used as a corrective measure on a defective Lot or Sub-Lot, the overlay thickness will be subject to the acceptance of the Ministry Representative, but shall not be less than 40 mm. In all other respects, the whole overlay will be subject to the same specifications as the pavement being overlaid. Where an overlay is used as a corrective measure in any lane, adjacent lanes shall also be overlaid whether acceptable or not.

Whether the overlay is applied as a corrective measure or is placed over otherwise acceptable pavement in order to match an adjacent lane, acceptability and payment will be determined as follows:

(a) Acceptability, and eligibility for either positive or negative payment adjustment, will be determined entirely on the results of testing and observations conducted on the overlay, regardless of test results that have been obtained on the underlying, overlaid Lift of pavement; but

(b) The payment quantity, for application of the Unit Prices for asphalt pavement, and the quantity, to which any payment adjustment is to be applied, will be derived from the tonnage of mix in the underlying, overlaid Lift.

APPEAL TESTING

502.60 Time Limits for Appeal of Test Results – To appeal any test results, the Contractor shall serve written notice of appeal to the Ministry Representative, within the following period after receipt of the applicable test results:

(a) For Density, Gradation and Asphalt Content, within three (3) days; and

(b) For Segregation, Smoothness and Application Rate, within five (5) days.

502.61 Appeals of Density, Asphalt Content and Gradation – The Contractor may appeal the results of acceptance testing for density, asphalt content and/or aggregate gradation for any Lot only once. Appeals will only be considered if quality control results support an appeal and can be presented.

The Ministry Representative will arrange for an independent testing laboratory to perform the appeal testing. The personnel employed or testing laboratory retained by the Contractor for quality control testing on the project will not be used for appeal testing.

The appeal testing laboratory shall hold current certification from the Canadian Council of Independent Laboratories (CCIL) (http://www.ccil.com/) under the Asphalt Laboratory Certification Programs, and at least one technician in the asphalt laboratory shall hold current certification under the Asphalt Technician Certification Program.

502.61.01 Density Appeals – For density appeals, Quality Control test results for density which are provided to the Ministry Representative subsequent to the Contractor's receipt of the quality assurance test results for that Lot will not be considered (when evaluating evidence) for an appeal.

The appeal shall be for the failed sample(s) within the Lot, and there will be no appeal allowed for single tests within a Sub-Lot.

Any attempt to improve density on the appealed Sub-Lot after the Ministry Representative has tested the Sub-Lot for acceptance shall void the appeal and the original test results will apply.

(a) The Contractor shall, within three (3) days of filing the appeal and in the presence of the Ministry Representative, take five (5) core samples from random locations from a Sub-Lot;

(b) The Contractor shall then deliver new core samples to the Ministry Representative;

(c) The Ministry Representative shall deliver the core samples and the companion loose mix samples from the appealed Sub-Lots to the appeal testing laboratory;

(d) The appeal agency shall prepare new briquette densities from the previously taken companion samples as per APPENDIX 502-B;

(e) The appeal agency shall determine the Bulk Density (BD) / Maximum Theoretical Density (MTD) from the companion sample and the densities of the cores and report the results to the Ministry Representative and the Contractor;

(f) The original core test results will be discarded, and a new sample mean will be calculated from the five (5) random cores and shall be used for acceptance and payment adjustments for the Sub-Lot.

502.61.02 Asphalt Content and Gradation Appeals – For asphalt content and gradation appeals:

(a) The party who has possession of the companion loose mix samples shall deliver them to the appeal testing laboratory.

(b) The appeal agency shall prepare three (3) new samples from the companion sample and determine the AC contents and gradations, average the results, and report all results to the Ministry Representative and the Contractor.

(c) The original test results will be discarded. A new sample mean for the three (3) new test results will be determined using the companion samples and will be used for acceptance and payment adjustments.

502.62 Smoothness Appeals – The Contractor may appeal acceptance test results of a Lot only once.

The Ministry Representative will perform, and the Contractor will be given the opportunity to witness, the appeal testing, and the new results will be binding on the Contractor and the Ministry.

502.63 Segregation Appeals – Appeals of segregation ratings will be handled by a joint review with the Contractor. If consensus cannot be reached then the Ministry Representative shall engage the Joint Committee, comprised of representatives from the Ministry and B.C. Road Builders, to assess the area(s) in question.

502.64 Application Rate Appeals – Appeals of application rate will be handled by a joint review with the Contractor and Ministry Representative. If the issue cannot be resolved, the issue will be resolved through the dispute resolution provisions of the Contract General Conditions.

502.65 Application of Appeal Testing Results – The appeal test values, thus determined, in all cases, will be binding on the Contractor and the Ministry.

502.66 Payment for Appeal Testing – If the new results indicate a change in the payment adjustment in the Contractor's favour, then sampling and testing costs incurred during the appeal procedures for that Lot would be borne by the Ministry.

If the new results verify that, any payment reduction or rejection remains valid for that Lot, then the costs of testing (plus 10% mark-up) incurred during the appeal procedure will be charged to the Contractor.

APPENDIX 502-A

QUALITY CONTROL REQUIREMENTS AND GUIDELINES

502-A.01 General – The Contract requirements for preparing, submitting and adhering to the paving components of a Quality Control Plan are specified in SS 502.04. This APPENDIX 502-A titled "Quality Control Requirements and Guidelines" sets out the guidelines for the Contractor's Quality Control Plan, in addition to those set out in SS 502.04 and the Special Provisions**.**

502-A.02 Quality Control Plan – The Contractor shall prepare and submit a Quality Control Plan for review. A qualified Subcontractor or an independent organization/agency may operate the Plan wholly or in part. However, the Quality Control Plan, including compliance with the Plan and its modifications, must remain the responsibility of the Contractor.

502-A.02.01 Plan Elements – The Plan shall address all elements that affect the quality of the hot mix, hot laid asphalt pavement, including but not limited to the following:

(a) Purchased Materials (asphalt cement characteristics, additives, purchased aggregates, etc.)

(b) Aggregate production (including source quality, gradation, fracture, crushing procedures, stockpiling, etc.)

(c) Calibration and Correlation of Testing Equipment (plant sensors, lab equipment, nuclear moisture/density gauges, etc.)

(d) Mix Design

(e) Asphalt Plant Mixing procedures (cold feed sampling, AC flow rate, temperature control, records, weigh scale, etc.)

(f) Product quality (volumetrics, EPS payment elements, etc.)

(g) Professional standards (joints, placing temperature, rolling procedures, etc.)

502-A.02.02 Additional QC Plan Elements – The Plan shall also include the following:

(a) The name of the Quality Control testing agency and its proven capability to provide the specific services required for the project.

(b) The list of dedicated technical staff, if available, (including names, qualifications and relevant experience) and their proposed roles.

(c) The list of testing equipment available for project work.

(d) **Specific Personnel –** The Quality Control Plan shall include the designation of specific personnel to be responsible for specific quality control duties.

(i) **Quality Control Manager –** A Quality Control Manager (QCM) responsible for the development and management of the Quality Control Plan. This person shall be qualified as per the requirements identified in SS 502.04.03. This individual shall also be responsible for signing off Quality Control Testing and Inspection records. This individual shall also be responsible for ensuring the qualifications of quality control staff, implementing and documenting any changes or improvements to the Quality Control Plan.

(ii) **Process Control Technician –** There should be a designated Process Control Technician (PCT), with 5 years of related experience, who will ensure that laboratory test results and other quality control practices used to control the quality of aggregates and other mix components, and to adjust and control mix proportions to meet the mix design(s). The PCT is responsible for ensuring that testing equipment, utilized for proportioning and mixing are calibrated and in good working order. The Plan shall describe how the PCT's duties, including sampling methods and responsibilities are to be accomplished and documented. The Plan should also describe the criteria to be used by the PCT to correct or reject unsatisfactory materials.

(iii) **Pavement Quality Control Technician –** There should also be a Pavement Quality Control Technician (PQT), with five (5) years of related experience, who will ensure that delivered materials meet the requirements of the specifications. In addition, the PQT shall be responsible for periodically inspecting all equipment used in placing, finishing, and compacting to assure its proper operating condition and to assure that placing, application rate, finishing, joint construction, and compaction is in conformance with this specification and the contract requirements.

502-A.03 Contractor's Record of Quality Control Testing –Test results should be made on specified forms or charts immediately after completion of each test. These test results are to be made available to the Ministry Representative upon request.

Records of gradation control, both during aggregate production and during the asphalt mixing operation, should

be kept on the form H0295 - Mechanical Analysis of Aggregates (available from the Ministry Representative).

502-A.04 Material Application Rate – The Contractor shall control the Material Application Rate by monitoring the amount of asphalt mix delivered to the road against the area covered by checking the application rate minimally every ten loads.

The Contractor shall advise the Ministry Representative in writing on an ongoing basis of the application rate.

502-A.05 Density – The Contractor should take core samples to determine actual pavement density. At the start of paving, the Contractor should take a minimum of two pavement cores from each Sub-Lot. The Contractor may employ a nuclear densitometer (or moisture/density gauge) to ensure intermediate density control. Two nuclear densities may be determined for each Sub-Lot, based on job mix densities obtained from the most recent plant briquettes.

502-A.06 Other Quality Control Procedures – The Contractor may initiate other Quality Control procedures as necessary for ensuring production of a quality product and include them in the Quality Control Plan. Procedures may also be introduced after the start of work as necessary as amendments to the Quality Control Plan.

502-A.07 Quality Control Testing Frequency – Minimum test frequencies guidelines for Quality Control are described in Table 502-Q,on the following page.

These are the minimum frequencies, and the Contractor is responsible to assess the need to increase testing frequency, where aggregate source is not uniform or if any other condition exists that may warrant it. QC frequencies may be reduced below this level, subject to the Ministry Representative's authorization, should the Contractor's QC Plan be proven very effective.

Table 502-Q: APPENDIX 502-A – Guidelines for Minimum Test Frequencies

	ASTM Test	Minimum Frequency
Tests During Aggregate Production	C136, Dry Sieve Analysis of Aggregate C117 Sieve Analysis of Aggregates by Washing *(Modified for Field Lab with drying done over a hotplate or similar heating element)*	• Split Stockpiles: 1 for each stockpile for every 2 hours of production. • One main stockpile: for every 300 tonnes. • Blend Sand: 1 for every 100 tonnes during stockpiling. • Natural filler: 1 for every 50 tonnes during stockpiling.
	D5821 Determining the Percentage of Fractured Particles in Coarse Aggregate	Every second coarse aggregate sieve test.
	C117 Sieve Analysis of Aggregates by Washing *(Modified for Field Lab)*	One (1) per shift on reduced sample obtained from combined samples from the crusher.
Asphalt Products Tests	Tack and Prime	Contractor's option.
Tests During Asphalt Plant Mixing	C136, Dry Sieve Analysis of Aggregate	One (1) of combined aggregate (off the belt) every 300 tonnes.
	C566 & D2216, Moisture Content	• Aggregate: Two (2) tests per Lot • Asphalt mix: One (1) on first Sub-Lot and every second day.
	C117 Sieve Analysis of Aggregates by Washing *(Modified for Field Lab)*	One (1) per shift on reduced sample obtained from combined samples from the plant cold feed.
	D5581 Resistance to Plastic Flow of Bituminous Mixtures Using Marshall Apparatus or D6927 Marshall Stability and Flow of Asphalt Mixtures	One set of three briquettes for 1,200 tonnes or Lot, whichever is less.
	D6307 Asphalt Extraction, Ignition Method	One per Lot.
	D5 Penetration of Bituminous Materials	One per Manufacturer's Batch. Samples should be taken for every 3000 tonnes of mix production.
	D2171 Viscosity	Contractor's Option.
	D2041 Maximum Theoretical Density	One per Sub-Lot.
Test During Asphalt Paving for Density Testing	D5581 Resistance to Plastic Flow of Bituminous Mixtures Using Marshall Apparatus or D6927 Marshall Stability and Flow of Asphalt Mixtures	One 15 kg sample for every Sub-Lot or minimum one (1) per day for field testing.
	Core Samples	At start, two cores for each Sub-Lot. After rolling pattern established, only one core for each Sub-Lot. Core sizes shall be in accordance with Table 502-H.

APPENDIX 502-B

LOOSE MIX SAMPLES

APPENDIX 502-B1: OBTAINING AND PREPARING LOOSE MIX SAMPLES

502-B1.01 General

(a) Three (3) uncompacted mix samples per Sub-Lot shall be obtained by the Contractor in accordance with ASTM D979, APPENDIX 502-B2, and in the presence of the Ministry Representative or a delegate thereof. Samples shall be taken from the roadway after being laid by the paver, at locations and times chosen by the Ministry Representative. The first sample shall be used by the Contractor for Quality Control, the second sample by the Ministry for Quality Assurance, with the third retained by the Ministry for potential appeal testing.

(b) The sizes of the samples taken shall meet the requirements of ASTM D979, Table 1, *Guide for Estimating Minimum Sample Quantity*.

(c) The sample size shall be reduced to required laboratory sample size for Marshall and Superpave gyratory briquettes, asphalt content and hot mix gradation determination as outlined below:

(i) The sample shall either be reduced using a Riffle splitter or shall be quartered into four approximately equal portions. The two diagonally, opposite quarters shall be combined resulting in two samples. Identify and designate one of the samples as the Quality Companion Sample and set aside. Identify and designate the other resulting sample as the Quality Control Sample. Use the Quality Control Sample for testing.

(ii) The Quality Control Sample shall be weighed to ensure that the sample so obtained meets the minimum mass required for the ignition test. If the sample does not meet the minimum mass requirements, then the additional materials will be obtained and added to the Quality Control Sample from the Companion Sample. This will be achieved by quartering the Quality Companion sample and adding one quarter of the Quality Companion Sample to the Quality Control Sample.

(iii) This process is to be repeated for each sample to provide the Quality Assurance and Appeal Samples.

(iv) These processes are to take place in the presence of the Ministry Representative or delegate thereof, at the time and location where the samples are taken, or another nearby location acceptable to the Ministry Representative.

(d) The third sample for appeal purposes shall be set aside and retained in a Contractor-supplied, suitable container labeled with sample location, date sampled, and project information.

(e) The Contractor shall deliver the QA and appeal samples to the Ministry Representative for storing.

(f) The areas sampled shall be filled with mix immediately after sampling and shall receive the same compactive effort as the rest of the mat.

APPENDIX 502-B2: LOOSE MIX SAMPLING PROCEDURE

502-B2.01 General – The purpose of this document is to establish a standard procedure for sampling loose mix asphalt used for the Ministry's projects. The intent is to obtain a representative loose mix asphalt sample before the material is compacted and incorporated into the pavement.

Three methods are outlined as follows. Users can select a method that is best suitable for their operations and can obtain a representative loose asphalt mix sample based on the individual project conditions.

502-B2.01.01 Method A (sampling using shovel – no plate) – In this method a loose mix sample is obtained from the asphalt concrete mat behind the paver using a shovel. The sample is obtained before the start of compaction.

502-B2.01.02 Method B (sampling using sampling plate) – This method uses a sampling plate that is placed on the roadway prior to placement of asphalt mix from the paver. This method is suitable when the asphalt mix is placed on a base course surface to avoid sample contamination with underlying base material. This method is also suitable when asphalt mix is placed on a milled surface to avoid loss of fines while obtaining the sample.

502-B2.01.03 Method C (sampling using sampling template) – This method uses a sampling template that is placed on the roadway prior to compaction. The template is inserted through the depth of the asphalt mat.

502-B2.02 General Training/Experience – All samples shall be obtained by a person experienced with procedures for sampling and handling asphalt mix. This person shall have adequate knowledge and training related to safety practices to be followed on a paving project.

502-B2.03 Apparatus/Equipment

(a) Equipment Common to All Methods

(i) Measuring tape

(ii) Insulated gloves

(iii) Flat nose shovel with vertical sides (Recommended: 10" x 12" with 2.5" sides, overall length 5'). Users can choose a shovel size suitable for their operations.

(iv) Sample container – Cardboard boxes, metal containers, stainless steel bowls or other suitable containers

(b) Additional Equipment Needed

(i) Method A

(A) No additional equipment needed.

(ii) Method B

(A) Shovel – Regular shovel can be used as well for this method.

(B) Sampling plate – heavy gauge metal plate preferably 380 mm x 380 mm (15" x 15") or 360 mm x 720 mm (14" x 28"), minimum 8 gauge thick optional wire attached to one corner long enough to reach from the center of the paver to the outside of the farthest auger extension. Holes 6.4 mm (¼") diameter should be provided in each corner. The plate should be capable of safely holding 15 to-30 kg material.

(C) Flat Tray – Flat tray with handles (can be used instead of plates).

(iii) Method C

(A) Shovel – Regular shovel can be used as well for this method.

(B) Sampling Template – Formed steel box (cookie cutter) with two 100 mm x 150 mm x 9 mm handles, sized to accommodate sample requirements. The leading edge of the cookie cutter box can be slightly beveled or angled for easier insertion into the asphalt mat.

502-B2.04 Sampling Procedure:

(i) Method A – Sampling Using Shovel (No Plate):

(A) Select a random location that is representative of the cross section of the mat as per the practices followed by the Ministry.

(B) Mark the roadway with suitable marker. Wait until the paver has passed the mark.

(C) For a standard 3.65m wide mat, samples shall be taken 0.3 to 1.0m from outside edge of the pavement.

(D) Sampling shovel can be preheated prior to taking the sample. Make sure that the shovel is free of any material, contaminant or solvent (release agent etc.).

(E) Using the flat nose sampling shovel at the random location, dig directly downward into newly placed asphalt mat until it comes into contact with the pavement or other surface below.

(F) When in contact, push the shovel forward until sampling shovel is full of asphalt mix and lift the shovel up slowly, being carefl not to lose any asphalt mix. Place materials from shovel directly into sample container.

(G) Scrape any asphalt mix material adhering to the shovel into the sampling container using a spatula. Extreme care should be taken to minimize coarse and fine particle separation while the sample is being taken.

(H) The asphalt mix sample shall be obtained from the full depth of the mat avoiding contamination from underlying layers.

(I) Try not to disturb any more sampling area than necessary to obtain a representative sample.

(J) Take sufficient asphalt mix material to complete all the testing requirements specified as per the contract.

(K) The material removed from the roadway shall be immediately replaced. The material shall not be broadcasted across the asphalt mat.

(ii) Method B – Sampling Using Metal Plate:

(A) Select a random location that is representative of the cross section of the mat to be constructed, as per the practices followed by the Ministry.

(B) The plate should be placed at least 600mm away from the edge of the asphalt mat. Determine the location of the plate before the paver reaches the selected location.

(C) Lay the plate (with attached wire if used) down diagonally with the direction of travel, keeping it flat and tight to the base with the lead corner facing the paving machine.

(D) If multiple plates are used (usually 3), the plates should be staggered over the lane, in

the transverse direction (two plates at 300 mm from the edges of the lane and one at the center of the lane).

(E) If wire is used with the plate, run the attached wire perpendicular to the direction of the paver, beyond the farthest auger extension and beyond the paving width. The wire should be held to the ground to allow construction and paving equipment to pass over the wire and the plate.

(F) Make sure that the plate is placed in a manner or is secured properly so that it doesn't get dragged or lifted during placement of asphalt mix. This is important as the plate can potentially cause screed or other equipment damage when not secured or placed properly.

(G) High temperature heat resistant metallic tape may be used to hold the plate securely (in case plate is getting dragged or lifted). The wire can be held to the surface or secured using tape.

(H) After the placement of asphalt mix, locate the plate. If wire attachment is used, locate the plate by raising the wire.

(I) Raise the plate slightly using a shovel, taking care not to lose any of the asphalt mix from the plate. This will locate four corners of the plate.

(J) Using the flat nose sampling shovel, dig directly downward into newly placed asphalt mat until it comes into contact with the plate below.

(K) When in contact, push the shovel forward until sampling shovel is full of asphalt mix and lift the shovel up slowly, being careful not to lose any asphalt mix. Place materials from shovel directly into sample container.

(L) Scrape any asphalt mix material adhering to the plate and shovel into the sampling container using a spatula. Extreme care should be taken to minimize coarse and fine particle separation while the sample is being taken.

(M) If flat tray with handles is used instead of plates, lift the flat tray using sampling shovel to locate corners of the tray. Flat tray should be secured to the surface prior to placement of mix as described above.

(N) Vertically lift the tray laden with asphalt mix from the HMA mat by being careful not to disturb the mix at the edge of the tray. Place the entire asphalt mix sample in the container. Scrape any asphalt mix material adhering to the tray into the sampling container using a spatula.

(O) The material removed from the roadway shall be immediately replaced. The material shall not be broadcasted across the asphalt mat.

(iii) Method C – Sampling Using Template ("Cookie Cutter"):

(A) Select a random location that is representative of the cross section of the mat to be constructed, as per the practices followed by the Ministry.

(B) Mark the roadway with a suitable marker. Wait until the paver has passed the mark.

(C) Once the paver has passed the sampling location, immediately place the template on the location to be sampled. Push the template through the asphalt mat until it touches the underlying surface.

(D) Using a shovel/scoop, carefully remove all the asphalt mix from inside the template and place into the sampling container.

(E) The template shall be scraped as clean as possible to include the fines in the sample.

(F) A plate can be used similar to Method B above if contamination of asphalt mix sample from underlying base or loss of fines on a milled surface is a concern.

(G) If sampling plate is used, make sure that the template is smaller than the plate and can be easily accommodated inside the sampling plate (example 330mm template dimension for a 380mm sampling plate).

(H) Scrape any asphalt mix material adhering to the template into the sampling container using a spatula. Extreme care should be taken to minimize coarse and fine particle separation while the sample is being taken.

(I) The material removed from the roadway shall be immediately replaced. The material shall not be broadcasted across the asphalt mat.

502-B2.05 Filling Sampling Holes:

(a) The holes shall be filled with sufficient quantity of asphalt (preferably slightly overfilled) so that satisfactory restoration of final surface can be achieved.

(b) When backfilling, the asphalt mix should be dumped vertically. Horizontal dumping of asphalt mix can result in segregation on the asphalt mat.

502-B2.06 Sample Identification / Labelling – Follow regular Ministry procedures for sample identification and labeling.

502-B2.07 Safety – Safety precautions are required when sampling and handling asphalt materials. Following are the general safety precautions to be observed when sampling. This procedure may not address all the safety concerns and additional safety measures may be necessary for individual projects or locations.

(a) Assessing hazards with associated work.

(b) Determining and using appropriate hazard mitigation strategies, to remove or eliminate hazards and prevent worker exposure.

(c) Taking precautions to avoid personal injury and environmental hazards.

(d) Taking precautions to prevent direct contact with high temperature materials.

(e) Using available resources for any emergencies.

(f) For safety, the roller(s) must remain at least 3 m (10 ft.) behind the sampling operation until the sample has been obtained and the hole filled with loose asphalt mixture.

(g) The plate placing operation must be at least 3 m (10 ft.) in front of the paver or pickup device. The technician placing the plate must have eye contact /communication with the paving machine operator. If eye contact cannot be maintained at all times, a third person must be present to provide communication between the operator and the technician.

(h) Specific precautions as required for each individual project and site.

502-B2.08 References:

(a) ASTM D979

(b) AASHTO R 97

(c) Elseifi, M., Evaluation of Hot Mix Asphalt Sampling Techniques (2007)

(d) Iowa DoT Hot Mix Sampling Demo (2019) [Source of the photos on the following pages.]

Recommended Sampling Shovel





Method A: Sampling Using Shovel Only



A1 - Digging Shovel into asphalt at selected location.



A2 - Lifting shovel full of asphalt mix.



A3 - Scraping off leftover asphalt mix/fines from the shovel into sample container.

Method B: Sampling Using Sampling Plate and Shovel



B1 – Placing sampling plates on the existing surface.



B2 - Plates prior to placement of asphalt mix by paver.



B3 - Locating sampling plate with attached wire after asphalt mix is placed.



B4 - Lifting sampling plate with shovel.



B5 - Taking mix sample on top of the sampling plate.



B6 - Filling the hole left behind by vertically dumping new asphalt mix.



B7 - Sampling hole filled up with new asphalt mix.



B8 - Sampling location after placement of new mix and compaction.

Method C: Sampling Using Sampling Template (Cookie Cutter)



C1 - Example: Sampling templates and scoop.



C2 - Placing sampling template at selected location on newly placed asphalt mat.



C3 - Placing sampling template at selected location on newly placed asphalt mat.



C4 - Removing asphalt mix inside the sampling template using scoop.